Customer Segmentation using RFM & Machine Learning

End-to-End Analytics Case Study
SQL → Power BI → Machine Learning → Streamlit

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SUMMARY

PROBLEM

Businesses lack visibility into customer behavior, leading to inefficient marketing strategies, poor retention, and missed revenue opportunities.

Need: A structured system to segment customers and enable targeted decision-making.

APPROACH

Pipeline:
SQL → Power BI → Machine Learning → Streamlit

SQL: Data preparation
Power BI: RFM segmentation
ML: Hidden pattern discovery
Streamlit: Interactive insights

KEY INSIGHTS

- Revenue is concentrated among a small segment of customers
- Loyal customers form the core business base
- At-risk customers show declining engagement
- Rule-based segmentation misses hidden patterns

ML VALUE ADD

KMeans clustering identified misclassified customers and uncovered hidden patterns, enhancing segmentation accuracy beyond rule-based approaches.

BUSINESS STRATEGY

Loyalty Programs → Retain valuable customers
Discount Strategy → Reactivate at-risk customers
Retargeting → Bring back inactive users

OUTCOME

Clear segmentation, improved targeting, reduced churn risk, and better decision-making.

FINAL INSIGHT

RFM provides clarity, ML adds depth → Combined approach gives optimal results.

1. Business Problem

In modern data-driven businesses, understanding customer behavior is critical for improving revenue, retention, and marketing efficiency.

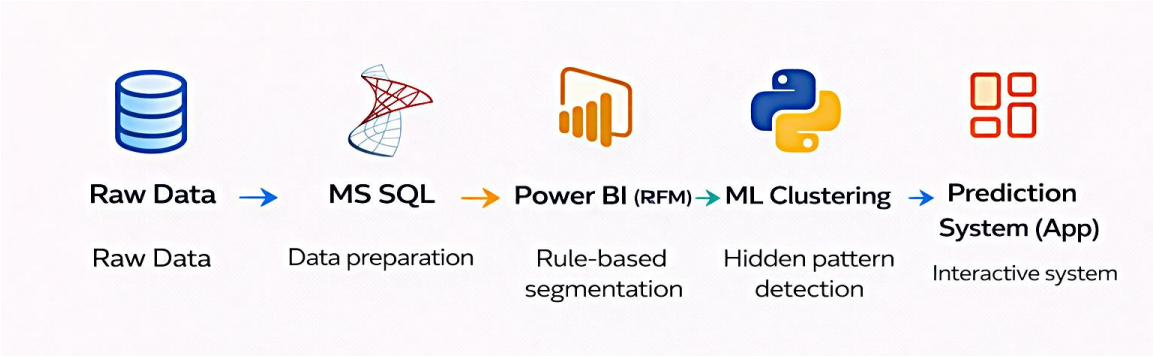
Challenges:
- Identifying high-value customers
- Detecting churn risk
- Applying targeted strategies

2. Objective

Build a customer segmentation system that combines business logic (RFM) with machine learning to improve decision-making.

3. Approach

Pipeline:
Raw Data → MS SQL → Power BI (RFM) → ML Clustering → Prediction System (App)



- SQL: Data preparation
- Power BI: Rule-based segmentation
- ML: Hidden pattern detection
- Streamlit: Interactive system

4. SQL Data Pipeline

Objective: Prepare structured customer-level dataset.

Note: No RFM segmentation applied here.

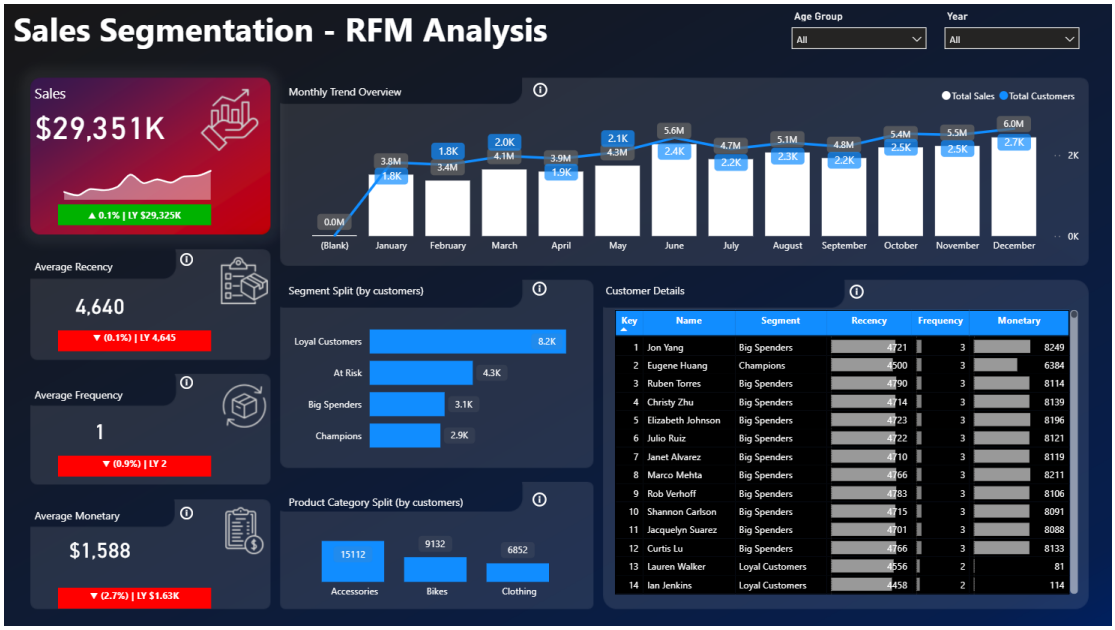
```sql
-- ==============================================================================
-- Create Report: vw_report_customers
-- ==============================================================================

IF OBJECT_ID('vw_report_customers', 'V') IS NOT NULL
    DROP VIEW [dbo].[vw_report_customers];
GO
CREATE VIEW [dbo].[vw_report_customers] AS

WITH base_query AS(
/*-----------------------------------------------------------------------------
1) Base Query: Retrieves core columns from tables
-----------------------------------------------------------------------------*/
SELECT
f.order_number,
f.product_key,
f.order_date,
f.sales_amount,
f.quantity,
c.customer_key,
c.customer_number,
CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
DATEDIFF(year, c.birthdate, GETDATE()) age
FROM [dbo].[gold.fact_sales] f
LEFT JOIN [dbo].[gold.dim_customers] c
ON c.customer_key = f.customer_key
WHERE order_date IS NOT NULL)

--SELECT * FROM base_query
```

Results | Messages

	customer_key	customer_number	customer_name	age	age_group	last_order_date	recency	total_orders	total_sales	total_quantity	lifespan	avg_order_value	avg_daily_spend
1	1	AW00011000	Jon Yang	55	50 and above	2013-05-03	4722	3	8249	8	8	2749	9
2	2	AW00011001	Eugene Huang	50	50 and above	2013-12-10	4501	3	6384	11	10	2128	6
3	3	AW00011002	Ruben Torres	55	50 and above	2013-02-23	4791	3	8114	4	4	2704	10
4	4	AW00011003	Christy Zhu	53	50 and above	2013-05-10	4715	3	8139	9	9	2713	9
5	5	AW00011004	Elizabeth Johnson	47	40-49	2013-05-01	4724	3	8196	6	6	2732	9
6	6	AW00011005	Julio Ruiz	50	50 and above	2013-05-02	4723	3	8121	6	6	2707	9
7	7	AW00011006	Janet Alvarez	50	50 and above	2013-05-14	4711	3	8119	5	5	2706	9
8	8	AW00011007	Marco Mehta	57	50 and above	2013-03-19	4767	3	8211	8	8	2737	10
9	9	AW00011008	Rob Verhoff	51	50 and above	2013-03-02	4784	3	8106	7	7	2702	10
10	10	AW00011009	Shannon Carlson	57	50 and above	2013-05-09	4716	3	8091	5	5	2697	9

Observations:
- Customer-level metrics created
- Recency, sales, and averages available
- Basic classification present

5. Power BI – RFM Segmentation



Contents

DAX USED IN Tbl_Measure Table 1
DAX for Creating Date Table 5
DAX for Creating RFM Calculation Table 6
DAX Calc Column within RFM Calculation Table 6

DAX USED IN Tbl_Measure Table

Total Customers = DISTINCTCOUNT("vw_reports_customers" [Customer ID])

Total Sales = SUM('vw_reports_customers'[Sales])

AvgRecency = AVERAGE(RFM_Calculation[Recency])

AvgFrequency = AVERAGE(RFM_Calculation[Frequency])

AvgMonetary = AVERAGE(RFM_Calculation[Monetary])

Segments:
Champions, Loyal, Big Spenders, At Risk, Lost

Segment	Recency (R)	Frequency (F)	Monetary (M)	Description
🏆 Champions	≤ 2	≤ 2	≤ 2	Best customers across all metrics
⭐ Loyal Customers	≤ 3	≤ 2	•	Frequent and recent buyers
💰 Big Spenders	≥ 3	•	≤ 2	High spenders but not recent
⚠️ At Risk	≥ 3	≤ 3	•	Previously active, now inactive
❌ Lost	= 5	= 5	= 5	Low engagement across all metrics
🔹 Others	•	•	•	Remaining customers

6. Business Strategy

2. Key Business Insightss

- Loyal Customers form the largest segment → strong retention base
- Big Spenders generate high revenue despite lower frequency
- At Risk customers show high recency → potential churn
- Champions are fewer but extremely valuable

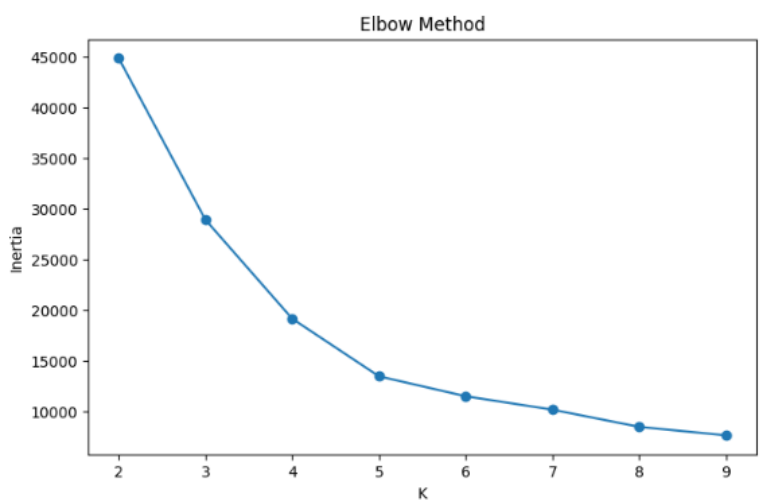
👉 Insight: Revenue is concentrated among high-value customers.

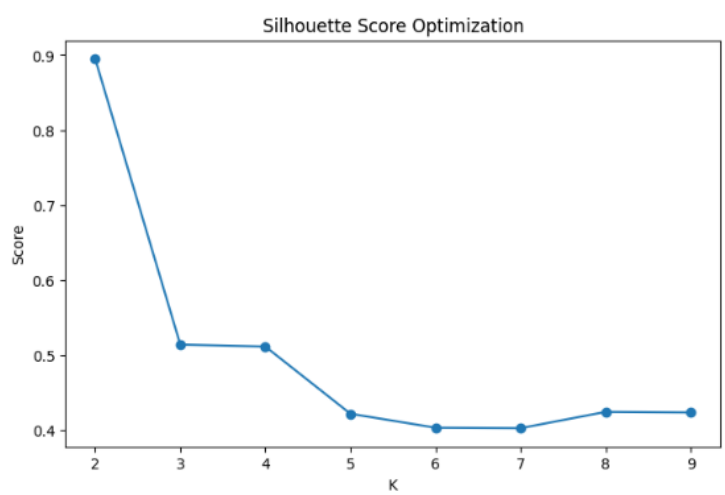
3. Strategy Summary Tables

Category	Loyalty Programs	Smart Discount Strategy	Retargeting Campaign
Target	Loyal Customers	At Risk + Big Spenders	At Risk + Lost
Why?	High frequency → engaged users Close to becoming Champions	High recency → not purchasing recently High value but low engagement	Low engagement Custome
Strategy	Rewards programs Membership tiers Early access	Personalized discounts Limited-time offers	Email campaigns Ads + no
Goal	Increase spending → convert to Champions	Reactivate customers before churn	Bring customers back

7. Machine Learning (KMeans)

Used clustering to identify hidden patterns.
Steps:
- Feature scaling
- Elbow method
- Silhouette score





```python
# ============================================================
# 🎯 SELECT BEST K
# ============================================================
optimal_k = results_df.loc[results_df['Silhouette'].idxmax(), 'K']
print(f"\n✅ Optimal K selected: {optimal_k}")
```

```
✅ Optimal K selected: 2
```

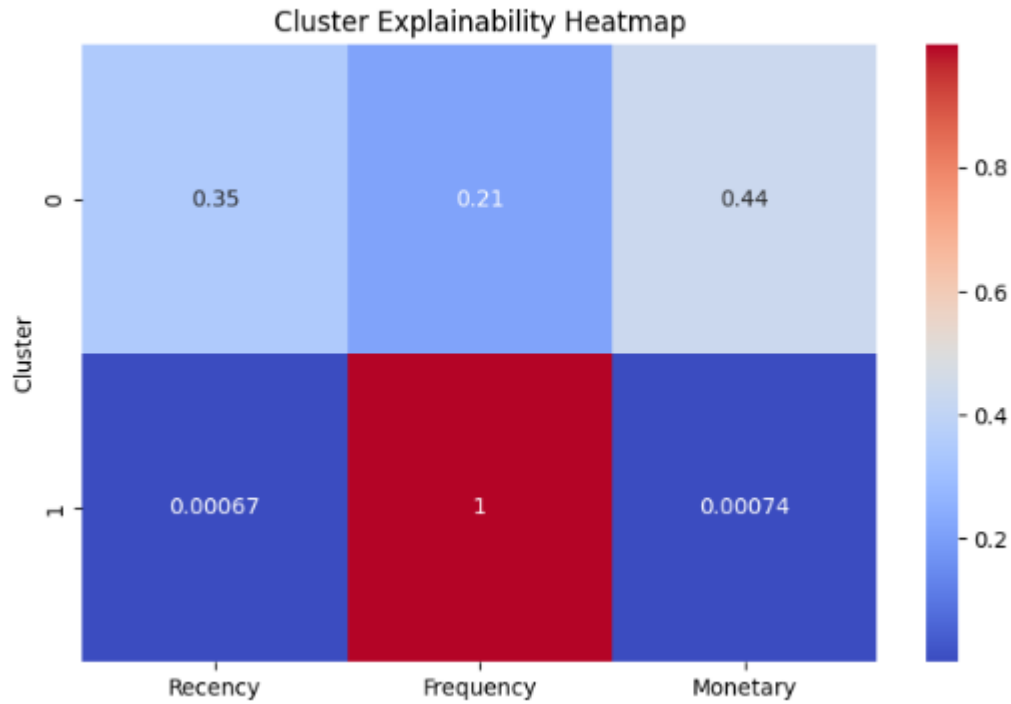
```python
# ============================================================
# 🚀 FINAL MODEL
# ============================================================
kmeans = KMeans(n_clusters=int(optimal_k), random_state=42)
df['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

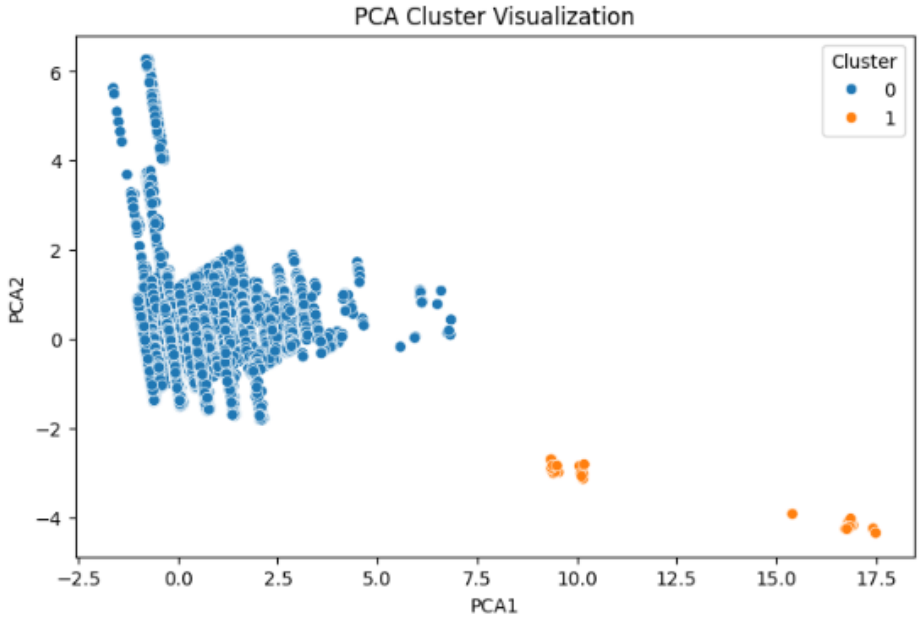
```python
# ============================================================
# 📊 CLUSTER SUMMARY
# ============================================================
cluster_summary = df.groupby('Cluster')[['Recency','Frequency','Monetary']].mean()
print(cluster_summary)
```

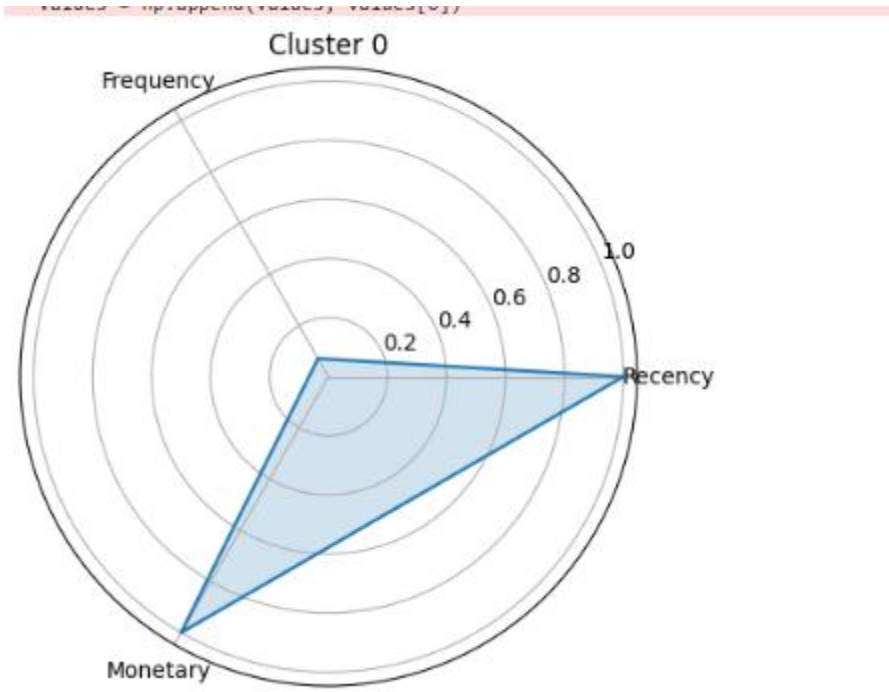
```
              Recency  Frequency     Monetary
Cluster
0        4640.661788   1.460183  1589.241340
1        4468.514286  20.600000   986.371429
```

```python
# ============================================================
# 🏷️ FINAL EVALUATION
# ============================================================
final_score = silhouette_score(rfm_scaled, df['Cluster'])
print(f"\nFinal Silhouette Score: {final_score:.3f}")
```

```
Final Silhouette Score: 0.895

 Cluster
 0        0.349057   0.214361  0.436582
 1        0.000666   0.998594  0.000740
```



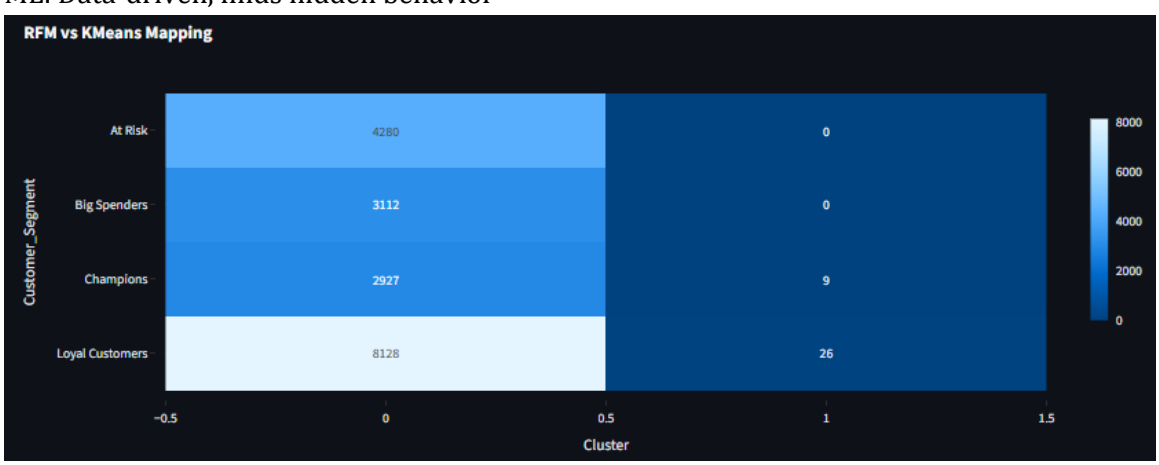




Observation:
ML reveals patterns missed by RFM

8. RFM vs ML Comparison

RFM: Rule-based, easy to interpret
ML: Data-driven, finds hidden behavior



Best approach: Combine both

9. Prediction System

Input customer metrics → Predict segment → Suggest strategy

Enables real-time decision-making

10. Conclusion

RFM provides clarity, ML enhances accuracy.

Combined approach improves targeting and business outcomes.